

Project Proposal: Watercolor Effect Using OpenCV and Neural Style Transfer

Team 7: Yu-Ting Cheng (ytcheng4), Zih-Yi Cao (zihyic2)

Motivation

Watercolor paintings, with their soft blends, smooth gradients, and expressive brushstrokes, present a fascinating challenge in image processing. This project aims to replicate these artistic effects digitally through three approaches:

- 1) OpenCV-based edge-preserving filters
- 2) Neural Style Transfer (NST)
- 3) a hybrid method combining both

By exploring these techniques. We hope to enhance our understanding of image filtering, deep learning models, and the potential benefits and limitations of integrating traditional computer vision methods with deep learning.

Milestones

Milestone	Description	Expected Completion
Implement OpenCV-based smoothing	Apply Bilateral Filter to smooth images, mimicking color blocks	Week 11
Implement edge detection	Extract image contours to simulate watercolor strokes	Week 12
Neural Style Transfer (NST) setup	Apply a pre-trained VGG-19 model and style transfer using watercolor paintings as style images	Week 13
Integrate and fine-tune the methods	Combine OpenCV smoothing with NST to refine and enhance results	Week 14
Evaluation and testing	Test on various images and analyze output quality	Week 15
Prepare final report and presentation	Compile findings and document the project	Week 16

Evaluation

We will visually compare the outputs from three methods: OpenCV-only, NST-only, and the hybrid approach. Evaluation will focus on how well each method replicates watercolor effects, including smooth color transitions and realistic brushstroke patterns. We will test the results across different image categories (e.g., portraits, landscapes) to assess overall effectiveness and versatility.

Resources

Pre-trained [VGG-19](#) model for NST (publicly available)

Python, OpenCV, and TensorFlow/PyTorch libraries

Watercolor artwork samples from Google or personal collections

Use Colab GPU for NST model

Group Contributions

Yu-Ting Cheng: Develop and implement the OpenCV-based method (Bilateral Filter, Edge Detection).

Zih-Yi Cao: Set up and fine-tune the NST model and style transfer pipeline.

Both: Collaborate on testing, final integration, and preparation of the report and presentation.